

Elijah Ballou

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Baltimore, MD

OBJECTIVE

Passionate game developer and UX/UI designer with a strong focus on Human-Computer Interaction (HCI), Virtual Reality (VR), Artificial Intelligence (AI), and Machine Learning (ML). I thrive on creating immersive gaming experiences and intuitive interfaces that merge cutting-edge technology with user-centered design. With a deep interest in AI-driven game mechanics, adaptive user experiences, and VR environments, I aim to craft innovative and engaging digital interactions. Seeking opportunities to apply my expertise in game development, AI, VR, and UX/UI to push the boundaries of interactive technology and create meaningful, next-generation experiences.

EXPERIENCE

- Noblis** May 2026 - August 2026
Unity VR Developer Intern Reston, VA
 - Develop immersive VR simulations and prototypes using Unity and C#.
 - Implement XR interactions, UI systems, and gameplay mechanics for Meta Quest.
 - Integrate emerging technologies into VR applications and research prototypes.
 - Test, debug, and optimize VR applications for performance and usability.
- Human-AI Experience (HAX) Lab** January 2023 - Present
Undergraduate Research Assistant Baltimore, MD
 - Develop research projects involving VR, game development, AI, and intelligent tutoring systems.
 - Design user interfaces and immersive interactions using HCI principles.
 - Prototype and enhance interactive technologies to improve system capabilities.
 - Conduct literature reviews and develop mockups to support research initiatives.
- Introduction to Game Design and Development** August 2023 - Present
Undergraduate Teaching Assistant Baltimore, MD
 - Lead hands-on labs focused on game development and Unity.
 - Assist students with C# scripting, debugging, and Unity development.
 - Manage course resources and provide technical support for student projects.
 - Grade projects and assessments according to established course criteria.

EDUCATION

- Morgan State University** January 2023 - Present
Bachelor of Science in Computer Science Baltimore, MD
 - GPA: 3.285/4.00

PROJECTS

- An Interactive Python Learning Platform: [Intelligent Tutoring System]** February 2025 - Present
Tools: [HTML/CSS, Python, PyTorch, MongoDB, Unity3D, C#] | Role: Team lead
 - Uses interactive challenges, gamification, and real-world projects to keep students motivated.
 - Adapts difficulty levels and content based on each student's progress and performance.
 - Uses points, badges, and leaderboards to keep students engaged and motivated while learning Python.
 - Includes assistive tools like text-to-speech, closed captions, and customizable UI to support diverse learners.
- VR Football Impact Simulator: [Head Injury Awareness]** August 2024 - Present
Tools: [Unity3D, Quest 2, C#, ConvAI, ElevenLabs] | Role: Lead Developer
 - Provides a first-person perspective of professional football gameplay, recreating the intensity of real-time collisions and their impact on players.
 - Explores the science behind concussions and chronic traumatic encephalopathy (CTE), educating users on the short- and long-term effects of repeated head trauma.
 - Challenges users with in-game scenarios that highlight the trade-offs between performance, safety, and career longevity in football.
 - Incorporates research-driven data on brain injuries, rehabilitation, and preventative measures, helping users understand the real-world implications of head trauma in sports.

• VR Survival Experience: Broadening Empathy, Agency, and Awareness

February 2024 – Present

Tools: Unity3D, Oculus Quest, C#, ConvAI | Role: Team Lead

- Developed an immersive VR survival experience modeling shelter access, food insecurity, environmental conditions, and social interactions faced by unhoused individuals.
- Designed choice-driven gameplay systems that emphasize player agency and illustrate how constraints and decisions impact short- and long-term outcomes.
- Integrated evidence-based information on homelessness and its societal causes within interactive mechanics to support learning through play.
- Implemented AI-driven conversational agents to simulate emotional and psychological challenges, fostering empathetic engagement and reflection.
- Led system design, technical integration, and overall project direction to ensure a cohesive and immersive user experience.

• Historical Pedagogical Agent Developer: [Immersive Experience of Public Figures]

March 2023 - Present

Tools: [Unity3D, C#, ConvAI, Blender, Render, ElevenLabs, SitePal] | Role: Team Member

- Players navigate shelter-seeking, food access, weather conditions, and social interactions, highlighting the obstacles faced by unhoused individuals.
- Integrates factual information on homelessness, its causes, and societal impact, encouraging players to learn while they play.
- Designed to broaden understanding and compassion for individuals facing homelessness by simulating their hardships and survival decisions.
- Experience the daily struggles of homelessness through interactive, choice-driven gameplay that reflects real-world challenges.
- Conversations with chatbots simulate the emotional and psychological challenges of homelessness, fostering empathy and deeper player connection.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [J.1] Ballou, E., Odunze, O., Adeleke, M. B., & Mack, N. A. **PyPro: Think in Code. Grow in Logic!**. In *Human-Computer Interaction & Emerging Technologies* 195 p.324. (2025)
- [J.2] Odunze, O., Ballou, E., Kanu, A., Kelly, A., Void, E., & Mack, N. A. **Inclusive Gaming Through Brain-Computer Interfaces: The Mind Mastery Experience**. In *Human-Computer Interaction & Emerging Technologies* 195, p.363. (2025)
- [C.1] Mack, N. A., Tandjong, C. W., Adeleke, M. B., Ballou, E., Harry, A., & Panton, J **Revolutionizing Culturally Relevant Math Education Through AI and Co-Designing Strategies**. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* pp. 1-6. (2025)
- [C.2] Mack, N. A., Adeleke, M. B., Ballou, E., Davis, D., Ingram, V., & Cox, K. **Breaking Stereotypes and Feeding the STEM Pipeline**. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1*, pp. 771-777. (2024)
- [C.3] Mack, N.A., Adeleke, M.B., Ingram, V., Ballou, E., Briggs-Belt, J.K. and Jordan, A. **CodeBears: Key Insights Gained from a Summer Coding Camp Empowering Underrepresented Youth**. In *Black Issues in Computing Education (BICE)*, pp. 80-86. IEEE (2024)
- [C.4] Meekins, S., Ballou, E., Mack, N.A. **Dreadphobia: Evaluating the Usability of a Virtual Reality Application in Support of Mental Health...** In *International Conference on Human-Computer Interaction*, pp. 397-402. Cham: Springer Nature Switzerland (2023)

SKILLS

- **Programming Languages:** C#, Python, JavaScript
- **Game & XR Development:** Unity, OpenXR, XR Interaction Toolkit, Meta Quest
- **Web Development:** HTML, CSS, React.js, Node.js, Flask
- **Database:** MongoDB
- **Version Control:** Git, GitHub
- **Development Design Tools:** Visual Studio Code, Blender, Figma, Canva
- **Technical Areas:** Interactive Systems, Human-Computer Interaction, Brain-Computer Interfaces

HONORS AND AWARDS

- **Morgan TechFest - 1st place winner** 2023
Morgan State University
 - Poster presentation competition
- **ACM Richard Tapia Conference - 2nd place winner** 2023
2023 CMD-IT/ACM Richard Tapia Celebration of Diversity in Computing Conference
 - Poster presentation competition
 - Tapia scholarship recipient
- **Morgan State: Codelinc - 2nd place** 2023
Lincoln financial Group
 - 24 Hour Hackathon
- **HBCU 2023 GAME JAM - Most Technically Challenging** 2023
Spelman College
 - 24 Hour Hackathon

PROFESSIONAL MEMBERSHIPS

- **Alpha Sigma Lambda Continuing Education National Honor Society** February 2025 - Present
- **National Society of Black Engineers**, Membership ID: 491132 October 2024 - Present
- **The Association for Computing Machinery**, Membership ID: 3061316 October 2023 - Present